

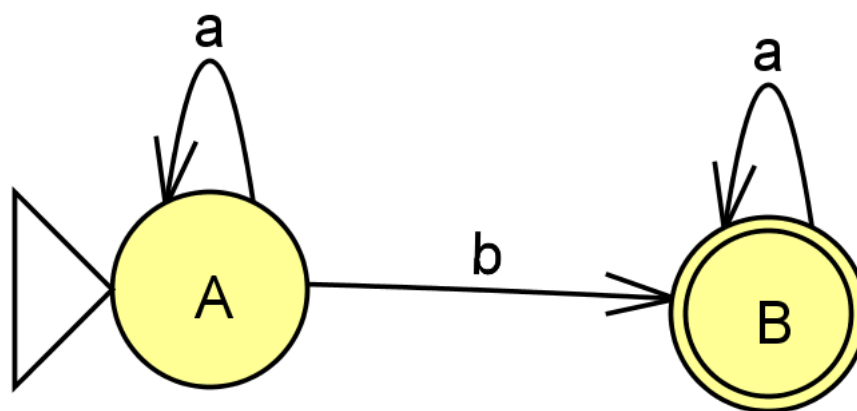
Deterministic Finite Automaton DFA

A DFA reads input and transitions from one state to another based on the symbol read from the input.

If the input ends and the DFA is in an accepting state, the input is accepted.

If the input ends and the DFA is in a state other than an accepting state, the input is not accepted.

Finite automaton, symbols a and b, the string must contain **exactly one b**



Initial state A reading aba ends in B accepting state => Input accepted

Example 1

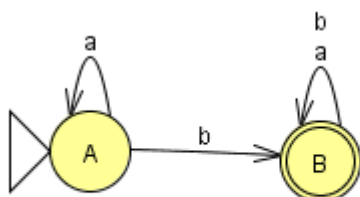
Automaton with letters (alphabet) a and b. Accepts a word if it contains (at least) one bb

aaaaab

aaabbb (???)

aaaaabbbaa (?)

Automaatti, jossa yksi tai useampi b



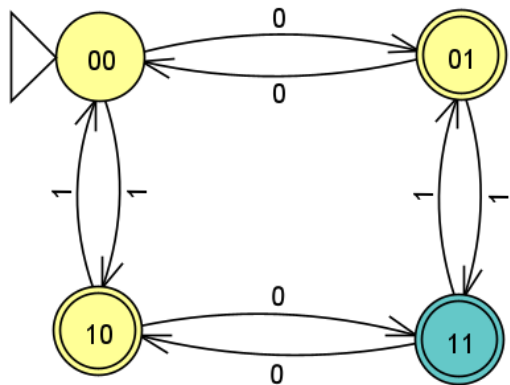
In the example done in the lecture (below), an empty string is also acceptable.

TBD

Example 2

As an example, consider an automaton that accepts a bit string $x \in \{1,0\}^*$ such that either the number of ones is odd or the number of zeros is odd.

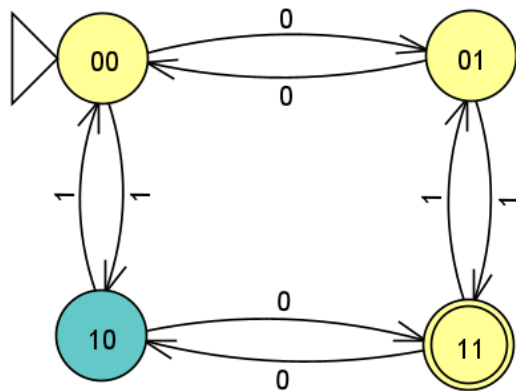
State	1 (ones)	0 (zeros)
00	0 even	0 even
10	1 odd	0 even
01	0 even	1 odd
11	1 odd	1 odd



Input	Result
01	Accept
00	Reject
11	Reject
10	Accept
101	Accept
110	Accept
000	Accept
1100	Reject
1000	Accept

Example 3

Let's change the previous one so that both must be an odd number



Input	Result
01	Accept
00	Reject
11	Reject
10	Accept
101	Reject
110	Reject
000	Reject
1100	Reject
1000	Accept

Let's take as an example an automaton that accepts a bit string $x \in \{1,0\}^*$ such that either the number of ones is even or the number of zeros is even

Quickly about regular expressions

- * zero or more: a^* can be a , aa , aaa , $aaaa...a$ or empty
- + one or more: a^+ can be a , aa , aaa , $aaaaa...a$ but not empty
- ? One or none: $a?$ is either a or empty
- Parentheses denote the scope of effect $(ab)^+$ can be ab , $abab$, $ababab...ab$

